Table summarizing the observed statistics for each benchmark program:

Benchmark	Instructions Executed	Cycles Taken	IPC (Throughput)
Descending	277	16062	0.01724567
Even or Odd	6	254	0.023622047
Fibonacci	78	3859	0.02021249
Palindrome	49	2338	0.020958085
Prime	29	1392	0.020833334

COMMENTS ON OBSERVATIONS ON DIFFERENT TYPES OF BENCHMARKS (Cycles Taken & IPC)

● **Cycles Taken:**

Across all benchmarks, the total number of cycles taken is much higher than the number of instructions executed. This mainly happens because of:

- The modeled latencies of memory and functional units, particularly the high latency of main memory.
- Data hazards that cause stalls in pipeline stages such as OF.
- Instructions that are fetched along incorrect branch paths and later squashed.

These factors introduce additional delays, which increase the overall number of cycles required for the simulation.

● **IPC (Throughput):**

The IPC values are low for all benchmarks, significantly lower than those observed in Assignment 4 where the pipeline model was more ideal.

This occurs due to:

- The latency-aware execution model used in the discrete event simulation.
- Increased waiting time caused by memory accesses and multi-cycle functional units.
- Pipeline stalls resulting from data hazards and instruction dependencies.

Among the benchmarks, Even or Odd achieves the highest IPC because it contains fewer instructions and experiences fewer stalls. On the other hand, Descending shows the lowest IPC, indicating heavier usage of memory operations and branch instructions.